CSA07 07

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SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. *(BL3)*

Assessment Tool Weightage: 30%

Code of Conduct:

I GIEONA ALIS. J (192511058) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

10marks Q5

Criteria	Weightage (%)	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks
Understanding of Scenario	20%				
Identification of Key Issues	20%				
Application of Concepts	25%				
Decision Making	20%				
Clarity of Response	15%				

29/50

Annexure A

Scenario-Based

1. A newly established Smart Campus is deploying a secure network infrastructure for its academic and administrative departments. The network administrator has been allocated the IP address block 192.168.100.0/24. Based on the expected number of devices, the following subnet masks have already been assigned:

School of Computing → /25

School of Electronics → /26

School of Mechanical Engineering → /27

As the network engineer, design the subnet allocation by performing the following tasks:

1. Determine the network address and broadcast address for each department.
2. Identify the valid host IP address range for each subnet.
3. Calculate the number of usable host addresses available in each subnet.
4. Determine the remaining unused IP address range within the given network block.

2. A multinational software company is expanding its headquarters and has been assigned the private IP block **10.10.0.0/16**. Different departments require different subnet sizes based on the number of employees. The network administrator has allocated the following subnet masks:

Software Development → **/24**

Quality Assurance → **/25**

Human Resources → **/26**

Executive Management → **/27**

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each department.
2. Identify the **valid range of host IP addresses**.
3. Calculate the **number of usable hosts** in each subnet.
4. Identify the **unused IP address range** remaining within the allocated IP block.

3.A smart hospital is migrating its network infrastructure from IPv4 to IPv6 to support thousands of connected medical devices and IoT sensors. The hospital has been assigned the IPv6 network:

2001:0db8:abcd:0000:0000:0000:0000:0000/64

As the network engineer responsible for the migration, perform the following:

1. Convert the given IPv6 address into its **compressed notation**.
2. Expand the compressed address back into its **full hexadecimal representation**.
3. Identify the **network prefix** using the **/64** prefix length.
4. Calculate the **total number of host addresses** supported within the subnet.
5. the total number of possible host addresses available within this network.

4. An Internet Service Provider (ISP) connects multiple customer networks through a core router. The router maintains the following routing table entries:

192.168.1.0/24

192.168.2.0/24

192.168.3.0/24

A packet arrives at the router with the destination IP address:

192.168.2.45

As the routing engineer, determine:

1. The **destination subnet** to which the packet belongs.

2. The **matching routing table entry**.
3. The **next-hop network/interface** used to forward the packet.
4. The **default route** that should be used if the destination does not match any routing table entry.

5. A company establishes two office departments connected through a Layer-3 router. The allocated subnet information is as follows:

Administration LAN → **192.168.10.0/26**

Finance LAN → **192.168.10.64/26**

The router interfaces are configured with:

Administration Gateway → **192.168.10.1**

Finance Gateway → **192.168.10.65**

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each LAN.
2. Identify the **valid host IP address range**.
3. Suggest suitable IP addresses for **three computers** in each LAN.
4. Identify the **default gateway** that hosts in each LAN should use for communication.

RUBRICS FOR CALCULATION

Scenario Based Assessment					
Criteria	**Weightage (%)**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Understanding of Scenario	20%	Demonstrates clear and complete understandin	Shows good understandi ng with minor gaps	Partial understandin g; misses key aspects	Misinterpret s or fails to understand the scenario

		g of the scenario and context			
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned , logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganize d and difficult to understand

QUESTION - 1

SMART CAMPUS - SUBNET ALLOCATION :

1) School of computing - 192.168.100/25

- Network Address : 192.168.100.0
- Broadcast Address : 192.168.100.127
- Valid Host Range : 192.168.100.1 - 192.168.100.126
- Usable host : 126

2) School of Electronics - 192.168.100.128/26

- Network Address : 192.168.100.128
- Broadcast Address : 192.168.100.191
- Valid host Range : 192.168.100.129 - 192.168.100.190
- Usable host : 62

3) School of Mechanical Engineering - 192.168.100.192/27

- Network Address : 192.168.100.192
- Broadcast Address : 192.168.100.223
- Valid host range : 192.168.100.193 - 192.168.100.222
- Usable host : 30

4) Remaining Unused IP Address Range

- Unused Network : 192.168.100.224 - 192.168.100.255

- Available for future Expansion : 32 IP addresses

CONCLUSION:

The subnet allocation efficiently divides the 192.168.100.0/24 network among the three departments while leaving

192.168.100.192 - 192.168.100.255

available for future use. This ensures efficient IP address utilization and supports network scalability

QUESTION - 2

MULTINATIONAL SOFTWARE COMPANY - SUBNET ALLOCATION :

1) Software Development - 10.10.0.0/24

- Network Address : 10.10.0.0
- Broadcast Address : 10.10.0.255
- Valid host Range : 10.10.0.1 - 10.10.0.254
- Usable hosts : 254

2) Quality Assurance - 10.10.1.0/25

- Network Address : 10.10.1.0
- Broadcast Address : 10.10.1.127
- Valid host range : 10.10.1.1 - 10.10.1.126
- Usable hosts : 126

3) Human Resources - 10.10.1.128/26

- Network Address : 10.10.1.128
- Broadcast Address : 10.10.1.191
- Valid host Range : 10.10.1.129 - 10.10.1.190
- Usable host : 62

4) Executive Management - 10.10.1.192/27

- Network Address : 10.10.1.192
- Broadcast Address : 10.10.1.223
- Valid Host Range : 10.10.1.193 - 10.10.1.222
- Usable host : 30

CONCLUSION:

The 10.10.0.0/16 network is efficiently divided into subnets based on department size. Each departments receives sufficient IP addresses, while the remaining address space is reserved for future growth, ensuring efficient network Management

QUESTION - 3

SMART HOSPITAL - IP V6 ADDRESSING

1) Compressed IPV6 Address

2001 : db8 : abcd ::/64

2) Expanded IPV6 Address

2001 : 0db8 : abcd : 0000 : 0000 : 0000 : 0000 : 0000/64

3) Network prefix

→ Prefix length :/64
→ Network prefix : 2001 : 0db8 : abcd : 0000 :: /64

4) Total host Addresses in the Subnet

→ Host bits : 64
→ Total host addresses : 2^{64} =
18, 446, 744, 073, 709, 551, 616.

5) Total possible host Addresses

- 18, 446, 744, 073, 709, 551, 616 (2^{64})

CONCLUSION:

The IPv6/64 network provides a very large number of host addresses, making it suitable for connecting thousands of medical devices and IOT sensors while supporting future network expansion.

QUESTION - 4

ISP ROUTING TABLE - PACKET FORWARDING

1) Destination Subnet

↑ Destination IP : 192.168.2.45

↑ Destination Subnet : 192.168.2.0/24

2) Matching Routing Table Entry

⇒ 192.168.2.0/24

3) Next - Hop Network / Interface

• The packet is forwarded through the interface connected to the 192.168.2.0/24 network.

4) Default Route

* If no routing table entry matches, the router uses the default route (0.0.0.0/0) to forward the packet to the next-hop router.

CONCLUSION:

The packet with destination 192.168.2.45 matches the 192.168.2.0/24 routing entry and is forwarded through the corresponding interface. If no matches exists, the router uses the default route (0.0.0.0/0) for packet forwarding.

QUESTION - 5

LAYER - 3 ROUTER - LAN CONFIGURATION:

1) Network Address and Broadcast Address

Administration LAN (192.168.10.0/26)

- Network Address : 192.168.10.0
- Broadcast Address : 192.168.10.63

Finance LAN (192.168.10.64/26)

- Network Address : 192.168.10.64
- Broadcast Address : 192.168.10.127

Valid LAN Host IP Address Range

Administration LAN

- 192.168.10.1 - 192.168.10.62

Finance LAN

→ 192.168.10.65 - 192.168.10.126

3) Suitable IP Address for three computers

Administration LAN

- PC1 : 192.168.10.2
- PC2 : 192.168.10.3
- PC3 : 192.168.10.4

Finance LAN

- PC1 : 192.168.10.66
- PC2 : 192.168.10.67
- PC3 : 192.168.10.68

CONCLUSION:

The two LANS are correctly configured with seperate /26 subnets. Each LAN has its own valid host range and default gateway, enabling efficient communication through the layer-3 router.